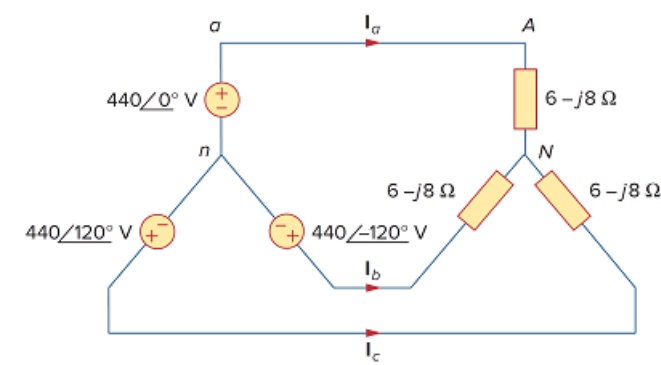


Problem

Obtain the line currents in the three-phase circuit of Fig. 12.42 on the next page.

Figure 12.42



Step-by-step solution

Step 1 of 1

This is a balanced Y-Y system.



Using the per-phase circuit shown above,

$$I_a = \frac{440\angle 0^\circ}{6 - j8} =$$

$$44\angle 53.13^\circ \text{ A}$$

$$I_b = I_a \angle -120^\circ =$$

$$44\angle -66.87^\circ \text{ A}$$

$$I_c = I_a \angle 120^\circ =$$

$$44\angle 173.13^\circ \text{ A}$$